Only in November 1915, after Einstein had formulated his general relativistic laws of gravity, did physicists once again believe they understood gravitation and light well enough to compute the effect of a star's gravity on the light it emits. Only then could they return with confidence to the dark stars (black holes) of Michell and Laplace.

The first step was made by Karl Schwarzschild, one of the most distinguished astrophysicists of the early twentieth century. Schwarz-schild, then serving in the German army on the Russian front of World War I, read Einstein's formulation of general relativity in the 25 November 1915 issue of the *Proceedings of the Prussian Academy of Sciences.* Almost immediately he set out to discover what predictions Einstein's new gravitation laws might make about stars.

Since it would be very complicated, mathematically, to analyze a star that spins or is nonspherical, Schwarzschild confined himself to stars that do not spin at all and that are precisely spherical, and to ease his calculations, he sought first a mathematical description of the star's exterior and delayed its interior until later. Within a few days he had the answer. He had calculated, in exact detail, from Einstein's new field equation, the curvature of spacetime outside *any* spherical, non-spinning star. His calculation was elegant and beautiful, and the curved spacetime geometry that it predicted, the *Schwarzschild geometry* as it soon came to be known, was destined to have enormous impact on our understanding of gravity and the Universe.

Schwarzschild mailed to Einstein a paper describing his calculations, and Einstein presented it in his behalf at a meeting of the Prussian Academy of Sciences in Berlin on 13 January 1916. Several weeks later, Einstein presented the Academy a second paper by Schwarzschild: an exact computation of the spacetime curvature *inside* the star. Only four months later, Schwarzschild's remarkable productivity was halted: On 19 June, Einstein had the sad task of reporting to the Academy that Karl Schwarzschild had died of an illness contracted on the Russian front.